

CLINICAL PRACTICE GUIDELINE: HYPERTENSION MANAGEMENT

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1. Diagnostic Criteria & Staging

- Normal Blood Pressure: Systolic < 120 mmHg and Diastolic < 80 mmHg.
- Elevated Blood Pressure: Systolic 120-129 mmHg and Diastolic < 80 mmHg.
- Stage 1 Hypertension: Systolic 130-139 mmHg or Diastolic 80-89 mmHg.
- Stage 2 Hypertension: Systolic $\geq$ 140 mmHg or Diastolic $\geq$ 90 mmHg.
- Hypertensive Crisis: Systolic > 180 mmHg and/or Diastolic > 120 mmHg.

2. Initial Non-Pharmacological Interventions

- Dietary Approaches to Stop Hypertension (DASH diet): High in vegetables, fruits, and whole grains.
- Sodium restriction: Target < 1,500 mg/day (or at least 1,000 mg/day reduction).
- Physical activity: 90-150 minutes of aerobic and dynamic resistance exercise per week.
- Moderation of alcohol intake and smoking cessation.

3. First-Line Pharmacotherapy

- Thiazide diuretics (e.g., Chlorthalidone 12.5-25 mg daily, Hydrochlorothiazide 25-50 mg daily).
- Angiotensin-Converting Enzyme (ACE) Inhibitors (e.g., Lisinopril 10-40 mg daily).
- Angiotensin Receptor Blockers (ARBs) (e.g., Losartan 50-100 mg daily).
- Calcium Channel Blockers (CCBs) (e.g., Amlodipine 5-10 mg daily).

Note: ACE inhibitors and ARBs should not be combined simultaneously due to renal risk.

CLINICAL PRACTICE GUIDELINE: TYPE 2 DIABETES MELLITUS

Document ID: MED-REF-2026-02 | Endocrinology Division

1. Glycemic Targets

- Target HbA1c for non-pregnant adults: < 7.0% (53 mmol/mol).
- Fasting and preprandial capillary plasma glucose: 80-130 mg/dL (4.4-7.2 mmol/L).
- Peak postprandial capillary plasma glucose: < 180 mg/dL (10.0 mmol/L).
- More relaxed targets (HbA1c < 8.0%) are appropriate for patients with severe hypoglycemia history.

2. Pharmacotherapy Pathway

- First-line therapy: Metformin (initial 500 mg twice daily with meals, titrating up to 2000 mg/day) combined with lifestyle intervention.
- In patients with established atherosclerotic cardiovascular disease (ASCVD), heart failure (HF), or chronic kidney disease (CKD):
 - * SGLT2 Inhibitors (e.g., Empagliflozin 10-25 mg daily, Dapagliflozin 10 mg daily).
 - * GLP-1 Receptor Agonists (e.g., Semaglutide 0.5-1.0 mg weekly, Liraglutide 1.2-1.8 mg daily).

3. Monitoring and Complication Screenings

- Annual comprehensive foot examination to evaluate neuropathy and peripheral vascular disease.
- Annual urine albumin-to-creatinine ratio (uACR) and estimated GFR.
- Dilated eye exam every 1-2 years by an optometrist or ophthalmologist.

CRITICAL PROTOCOLS: EMERGENCY RED FLAGS & CONTRAINDICATIONS

Document ID: MED-REF-2026-03 | Emergency & Critical Care

1. Acute Coronary Syndrome (ACS) Red Flags

- Crushing substernal chest pressure radiating to left jaw, neck, or shoulder.
- Associated diaphoresis, dyspnea, nausea, and lightheadedness.
- Immediate actions: Oxygen if SpO₂ < 90%, Chewable Aspirin 324 mg, Sublingual Nitroglycerin (if not contraindicated).
- Contraindications to Nitroglycerin: SBP < 90 mmHg, severe bradycardia, or PDE-5 inhibitor use within 24-48 hours.

2. Sepsis & Septic Shock (qSOFA Criteria)

- Respiratory rate ≥ 22 breaths per minute.
- Altered mental status (Glasgow Coma Scale < 15).
- Systolic blood pressure ≤ 100 mmHg.
- Management: Blood cultures prior to broad-spectrum IV antibiotics within 1 hour, 30 mL/kg crystalloid fluid bolus.

3. Anaphylaxis Emergency Action

- Sudden cutaneous symptoms (urticaria, angioedema) plus respiratory compromise or hypotension.
- Primary treatment: Epinephrine (1:1,000 / 1 mg/mL) 0.3-0.5 mg Intramuscularly into anterolateral thigh immediately.
- Never delay Epinephrine administration for secondary antihistamines or corticosteroids.